```
%root_path = fileparts(mfilename("fullpath"));
root_path = pwd;
data_path = fullfile(root_path,'sym4D_cutsAndFits');


if ~exist('cuts2fit200','var')
    ld = load(fullfile(data_path,'multicuts_fit_dataGH_ei200meV.mat'));
    cuts2fit200 = ld.cuts2fit200;
end
if exist('used_ds200','var')
    used_ds = used_ds200;
else
    used_ds = containers.Map()
    sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei200ref001off110_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir001')=sel_info;
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei200ref001off200_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir001')=sel_info;
    %
    sel_info.cuts = {cuts2fit200.cut200_sel};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_2peaks_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_2peaks')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};
    sel_info.en_range = 40:10:140;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei200Br200dir010_1peak_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir010_1peak')=sel_info;
    %
    sel_info.cuts = {cut(cuts2fit200.cut110_sel,[0,0.02,2],[])};
    sel_info.en_range = 30:10:110;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei200Br110dir010_2D';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir010')=sel_info;

    %----------------------------------------------------
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(1);
    sel_info.en_range = 30:10:110;
```

```
    sel_info.cut_dim  = 1;
    sel_info.fname='EnFit_Ei200ref001off110_1D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br110dir001_1D')=sel_info;
    %
    sel_info.cuts = cuts2fit200.cuts_dir001FFfit(2);
    sel_info.en_range = 30:10:130;
    sel_info.cut_dim  = 1;
    sel_info.fname='EnFit_Ei200ref001off200_1D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei200Br200dir001_1D')=sel_info;

    used_ds200 = used_ds;
end
```

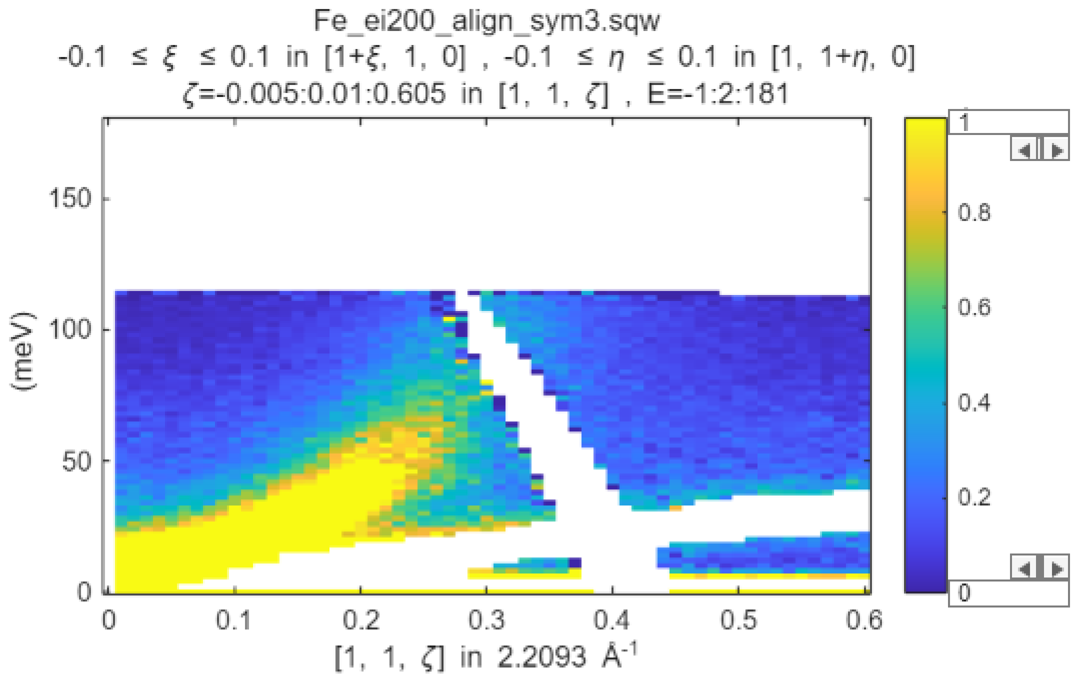
```
used_ds = 

  Map with properties:

        Count: 0
      KeyType: char
    ValueType: any
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  4690699 pixels -- processing data... -----> retained   4604422 pixels
--------------------------------------------------------------------------------
Number of points in input object: 8259853
     Fraction of object selected:    56.79% (=   4690699 points) ;  Access speed:     1.27GB/sec
     Fraction of object retained:    55.74% (=   4604422 points)

**** Total time in cut(sqw)   :      0.8sec Including:
     Data access time fraction:  32.9%; Transf. time frac.:  41.6%; Resulting pix preparation:  25.5%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  9054566 pixels -- processing data... -----> retained   8994792 pixels
--------------------------------------------------------------------------------
Number of points in input object: 10741088
     Fraction of object selected:    84.30% (=   9054566 points) ;  Access speed:     1.44GB/sec
     Fraction of object retained:    83.74% (=   8994792 points)

**** Total time in cut(sqw)   :      1.2sec Including:
     Data access time fraction:  34.2%; Transf. time frac.:  41.5%; Resulting pix preparation:  24.3%
--------------------------------------------------------------------------------
```

```

%cuts_list = cuts2fit200.cutsS_list();
%cuts_list = {cuts2fit200.cut200_200sel};
%cuts_list = {cuts2fit200.cut110_sel};

select= 'Ei200Br110dir001_1D';
sel_info = used_ds(select);

cuts_list = sel_info.cuts
```

```
cuts_list = 1×1 cell array
    {1×1 sqw}
```

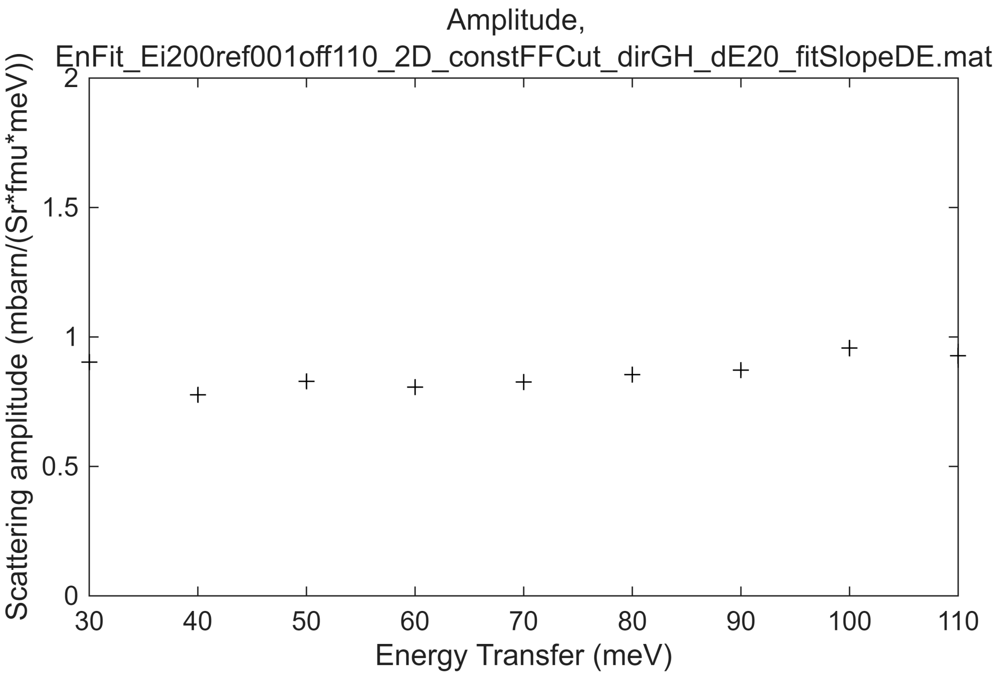
```
mi = maps_instrument(200,600,'S');
```

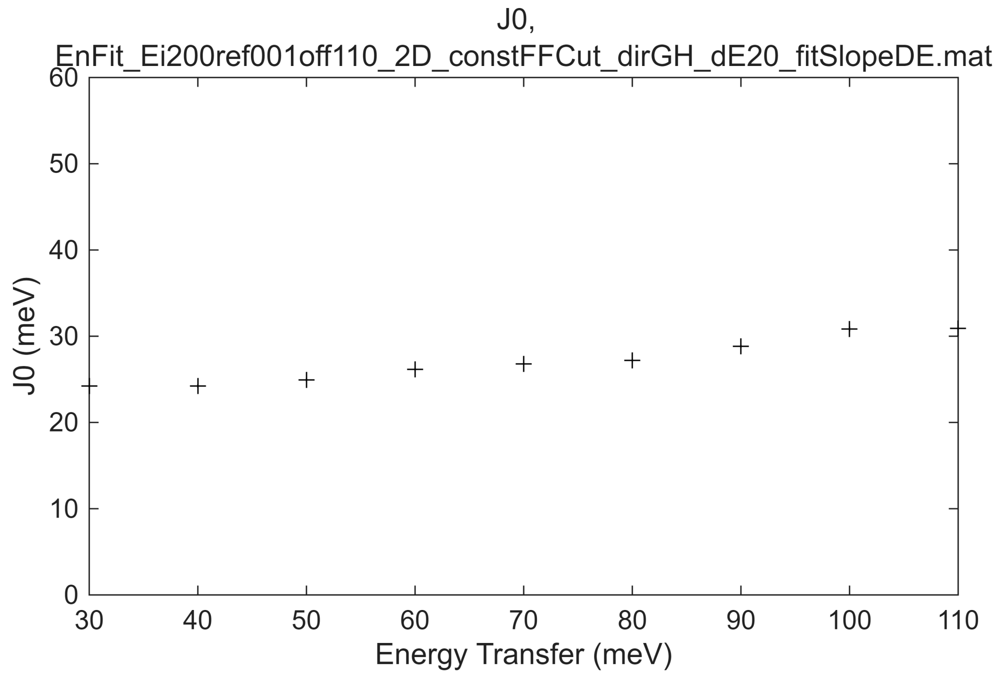
```
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end
```

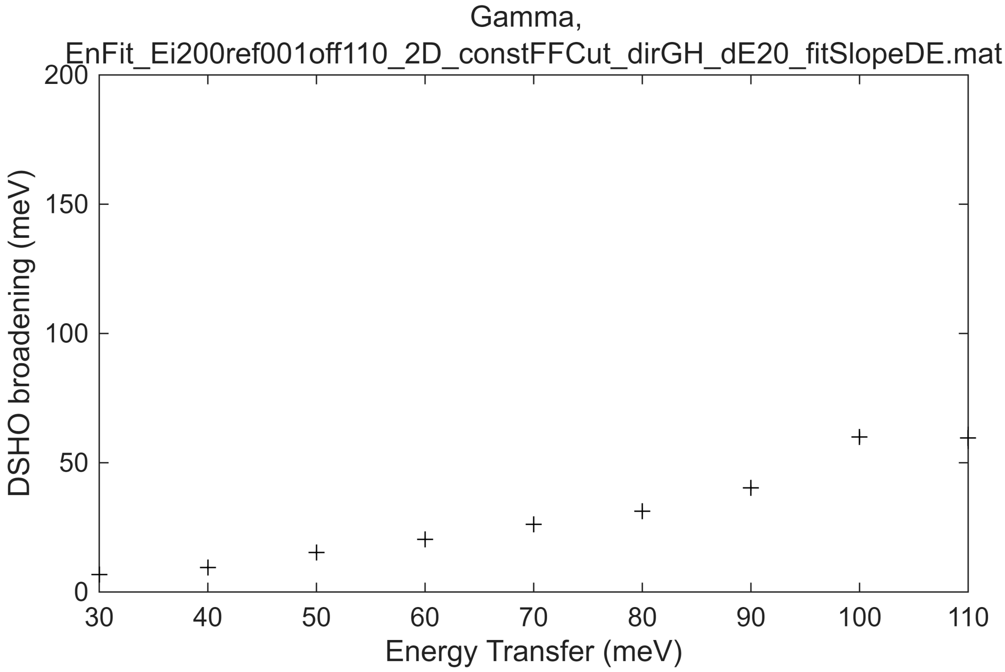



```
dir_name  = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 30:10:120;
%cut_en = 40:10:150;
%cut_en = 145;
cut_en = sel_info.en_range;
cut_dimensions = sel_info.cut_dim;

bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name  = sel_info.fname;
```

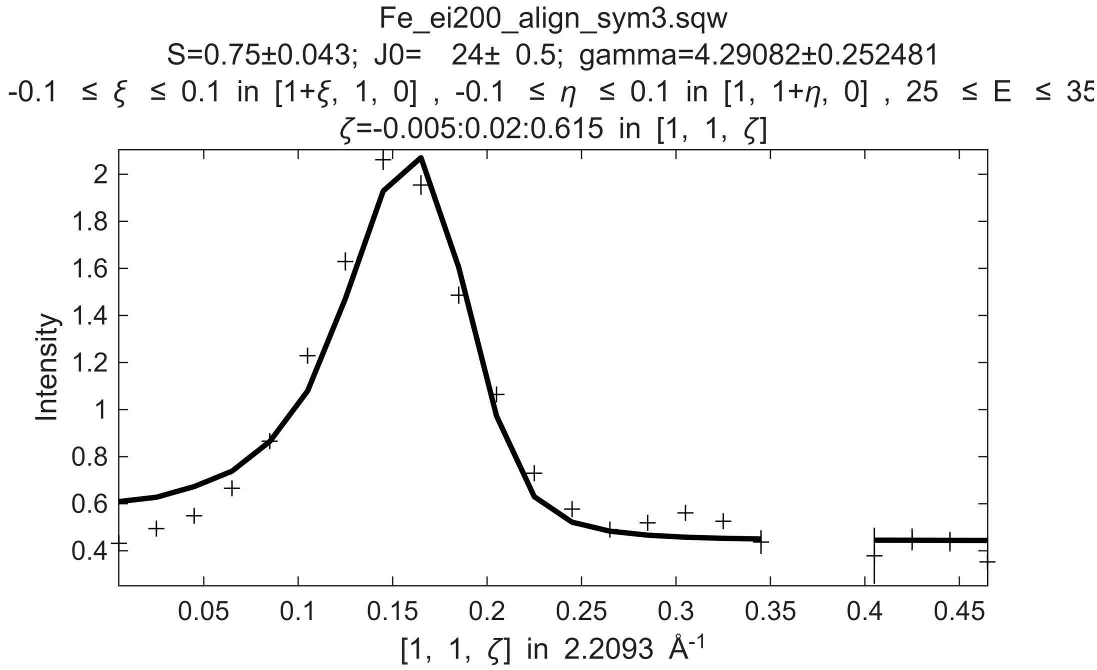
```
[fit_res_rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

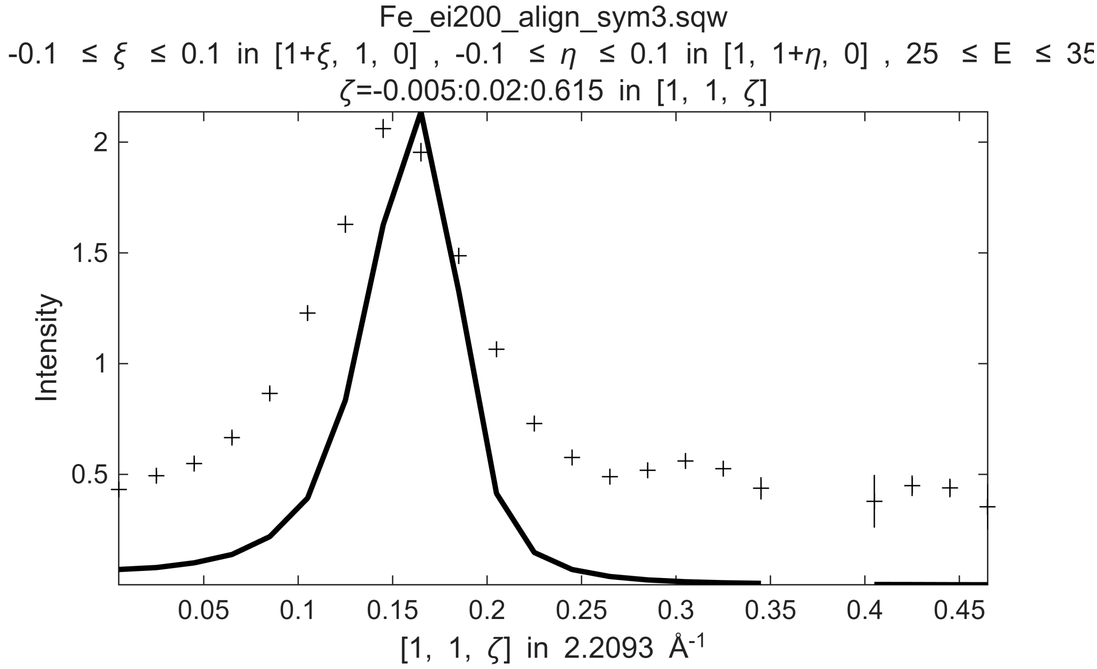

Amplitude,
EnFit_Ei200ref001off110_2D_constFFCut_dirGH_dE20_fitSlopeDE.mat
Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
0
0.5
1
1.5
2
30
40
50
60
70
80
90
100
110


J0,
EnFit_Ei200ref001off110_2D_constFFCut_dirGH_dE20_fitSlopeDE.mat
J0 (meV)
Energy Transfer (meV)
0
10
20
30
40
50
60
30
40
50
60
70
80
90
100
110


Gamma,
EnFit_Ei200ref001off110_2D_constFFCut_dirGH_dE20_fitSlopeDE.mat
DSHO broadening (meV)
Energy Transfer (meV)
200
150
100
50
0
30
40
50
60
70
80
90
100
110

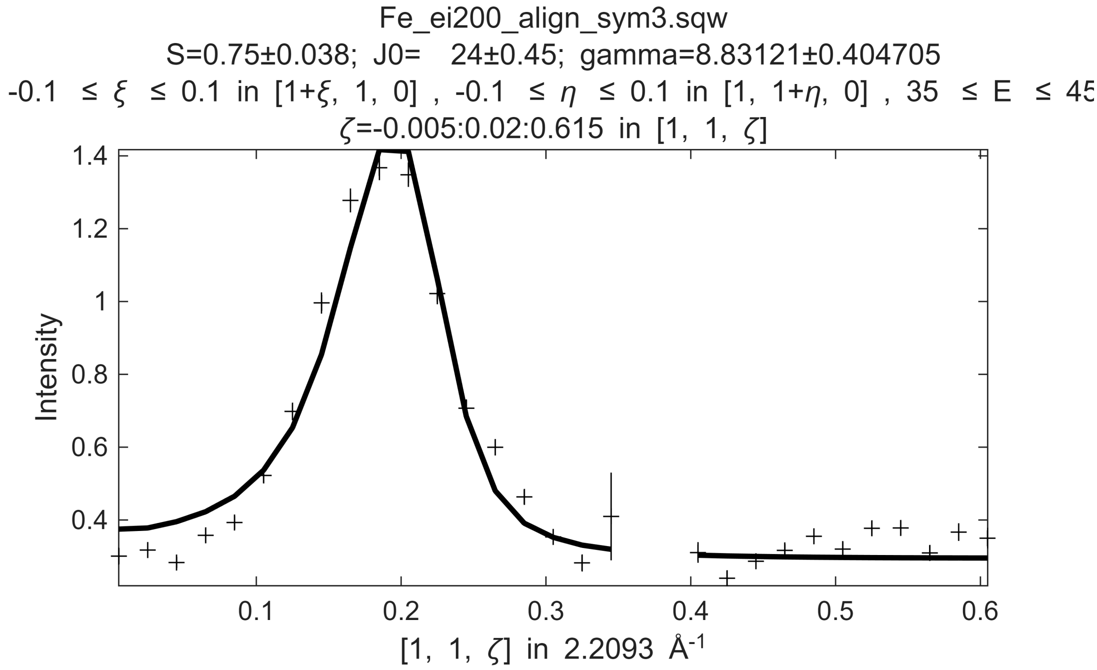
```
******************************
**** En = 30±5
******************************
```


Fe_ei200_align_sym3.sqw
S=0.75±0.043; J0=   24± 0.5; gamma=4.29082±0.252481
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 25 ≤ E ≤ 35
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

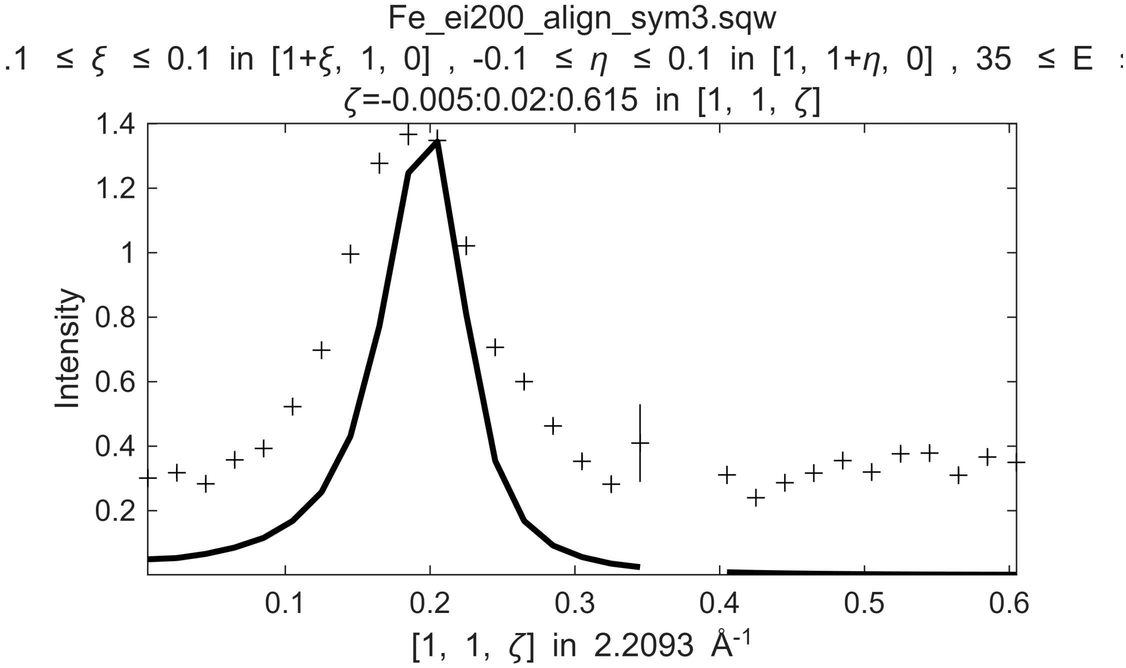


Fe_ei200_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 25 ≤ E ≤ 35
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

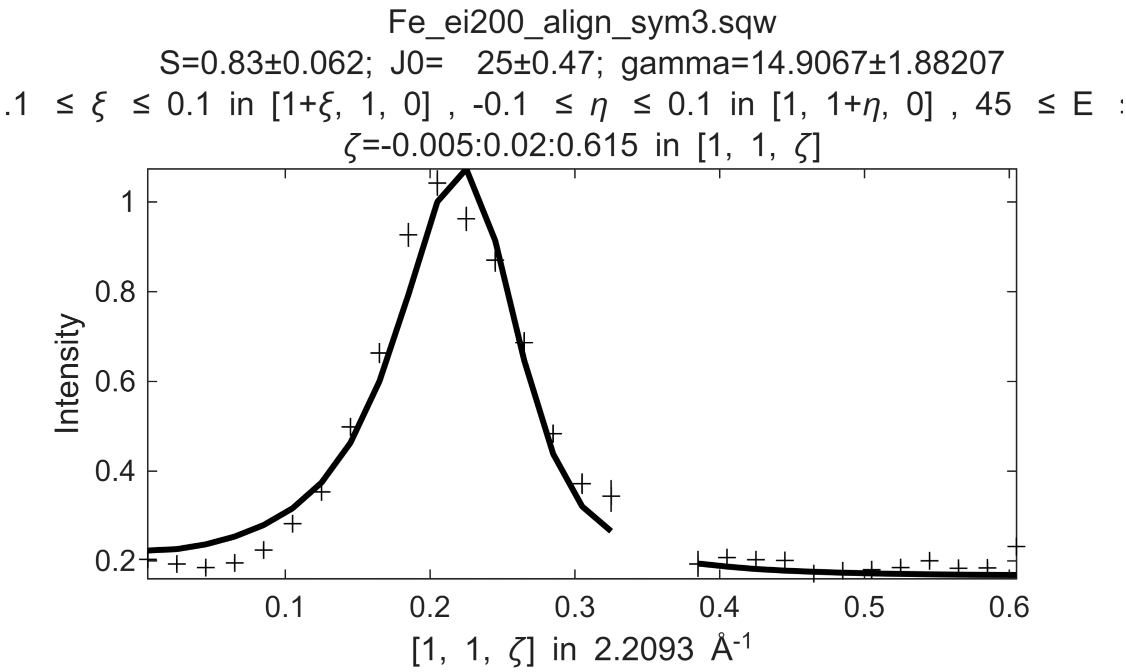

```
******************************
**** En = 40±5
******************************
```

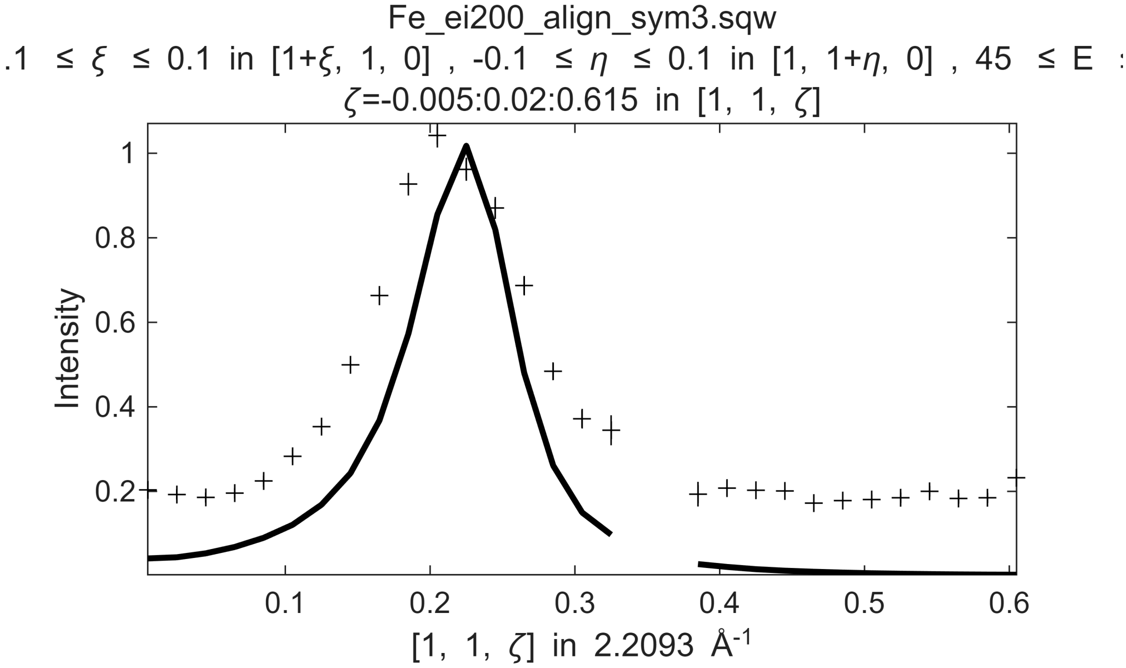


Fe_ei200_align_sym3.sqw
S=0.75±0.038; J0= 24±0.45; gamma=8.83121±0.404705
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 35 ≤ E ≤ 45
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

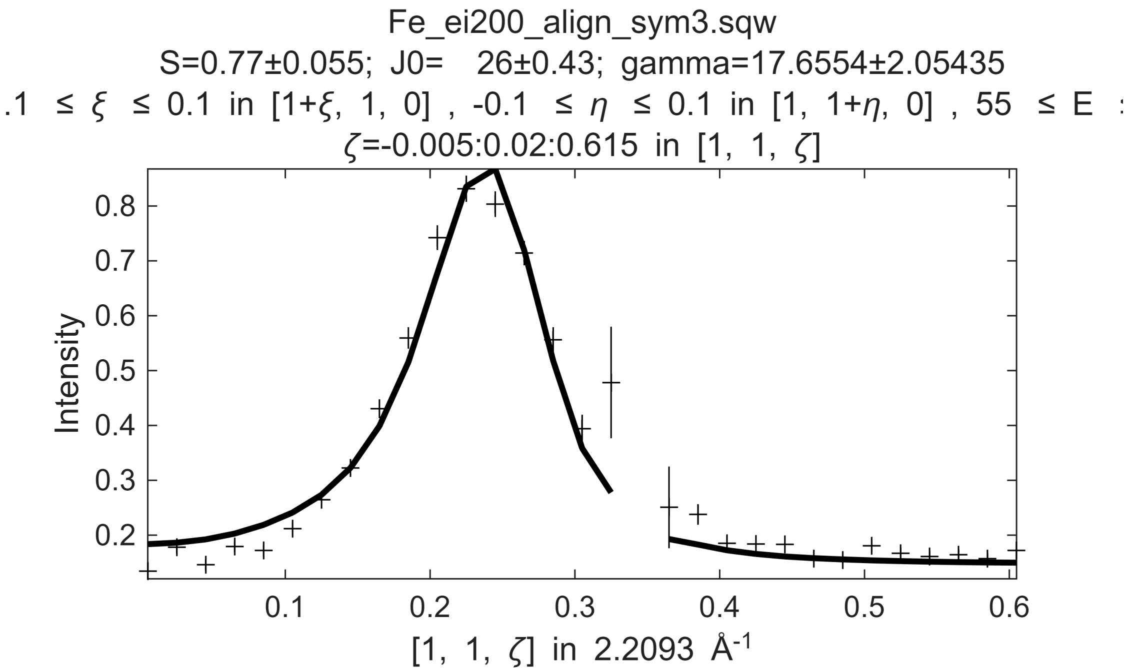

Fe_ei200_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 35 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

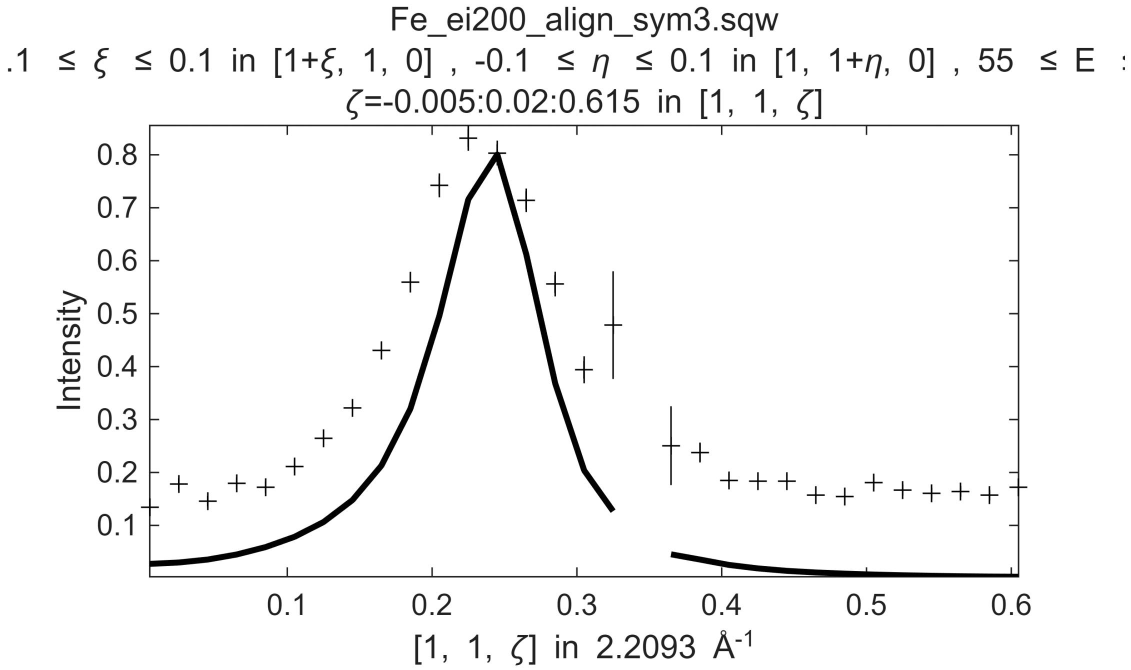
```
******************************
**** En = 50±5
******************************
```


Fe_ei200_align_sym3.sqw
S=0.83±0.062; J0= 25±0.47; gamma=14.9067±1.88207
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

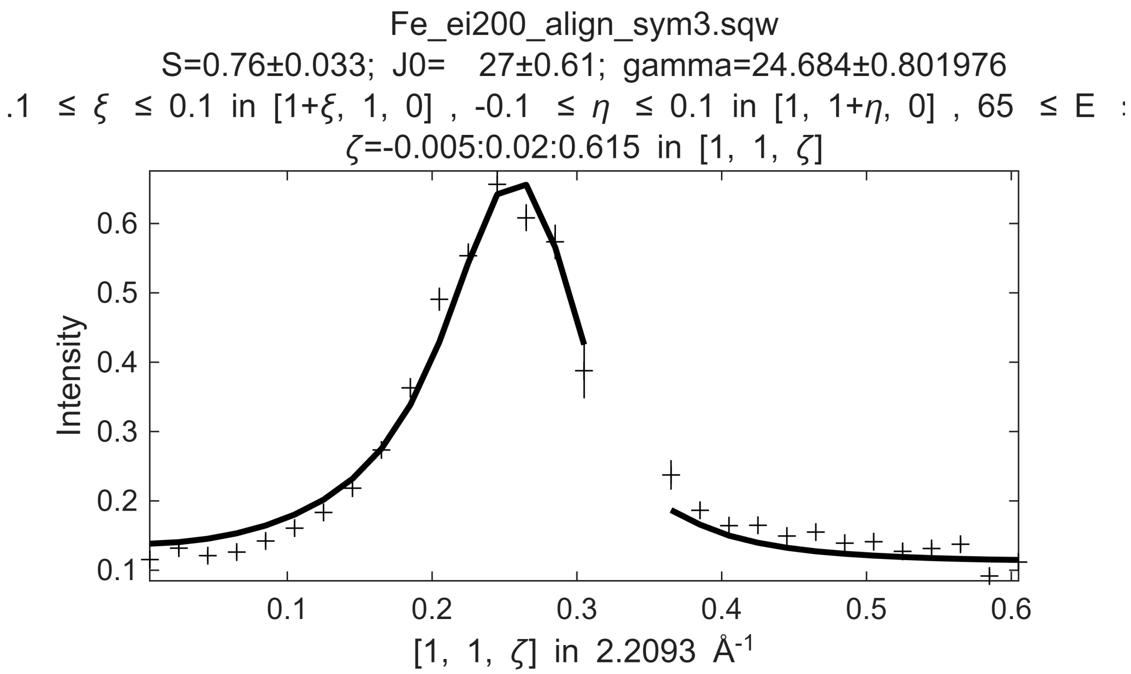

Fe_ei200_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å^-1

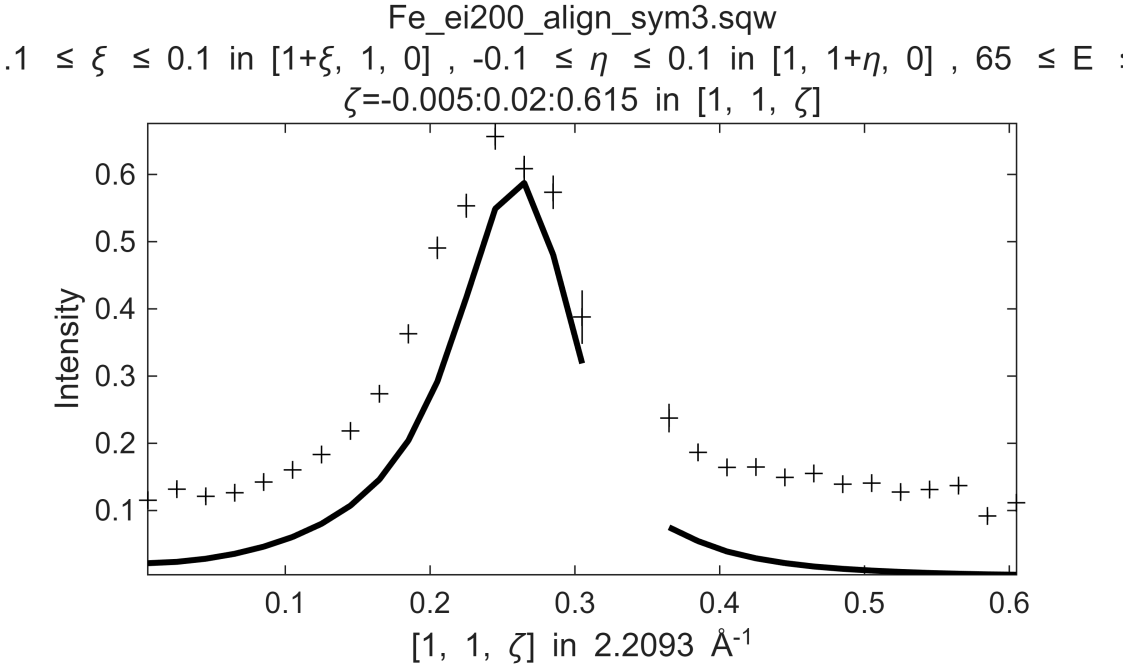
```
******************************
**** En = 60±5
******************************
```


Fe_ei200_align_sym3.sqw
S=0.77±0.055; J0= 26±0.43; gamma=17.6554±2.05435
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å^-1

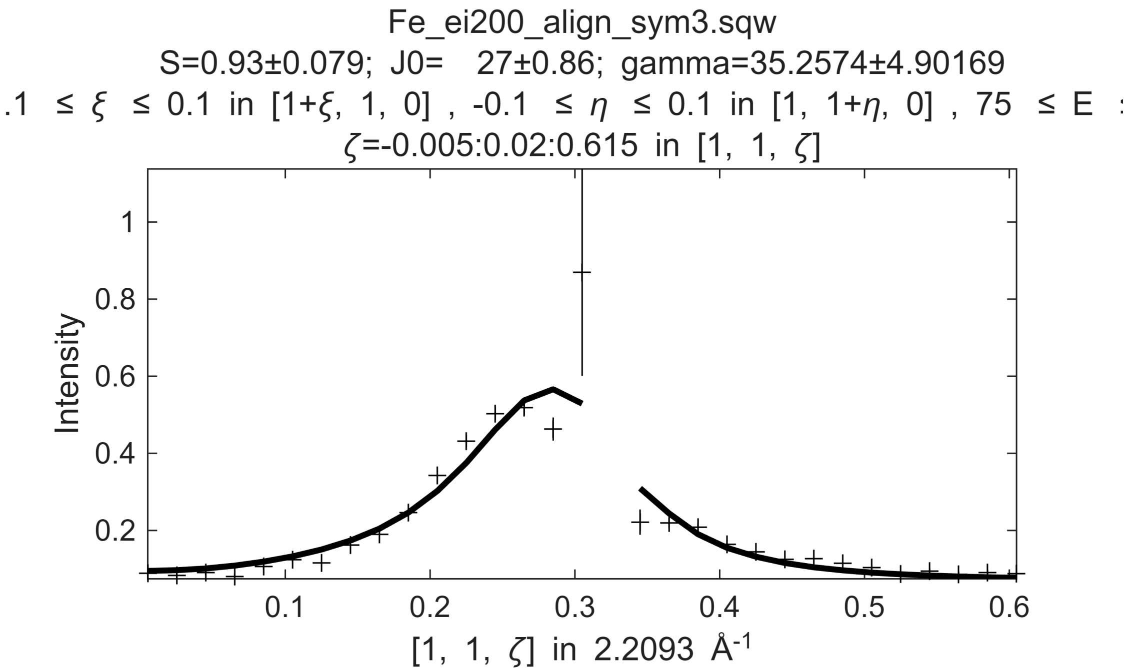

Fe_ei200_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å^-1

```
******************************
**** En = 70±5
******************************
```


Fe_ei200_align_sym3.sqw
S=0.76±0.033; J0= 27±0.61; gamma=24.684±0.801976
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å^-1


Fe_ei200_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

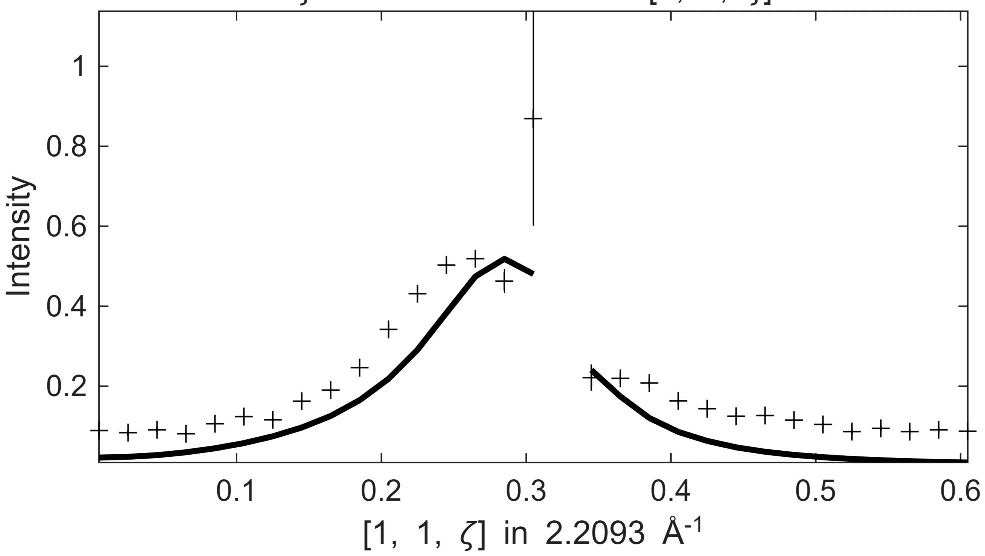
```
******************************
**** En = 80±5
******************************
```


Fe_ei200_align_sym3.sqw
S=0.93±0.079; J0= 27±0.86; gamma=35.2574±4.90169
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 75 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

Fe_ei200_align_sym3.sqw

.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 75 ≤ E

ζ=-0.005:0.02:0.615 in [1, 1, ζ]



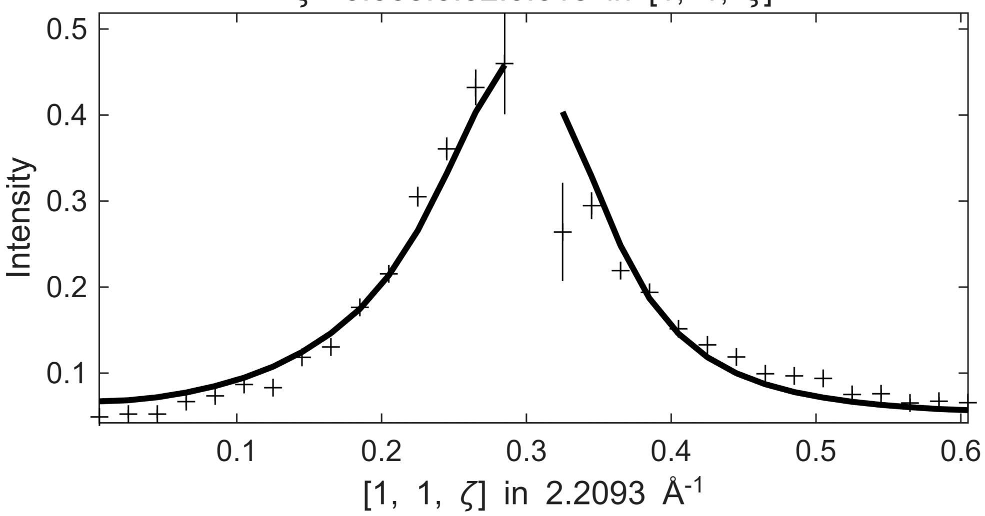

```
******************************
**** En = 90±5
******************************
```

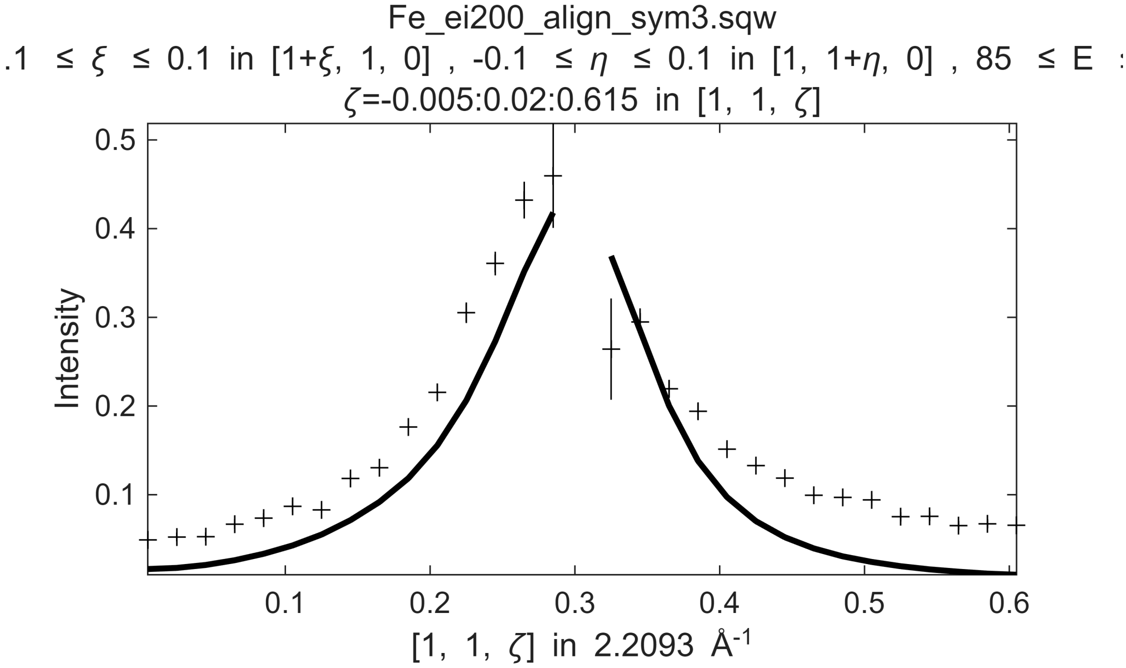
Fe_ei200_align_sym3.sqw

S=0.86±0.051; J0= 28±0.42; gamma=38.1565±2.46815

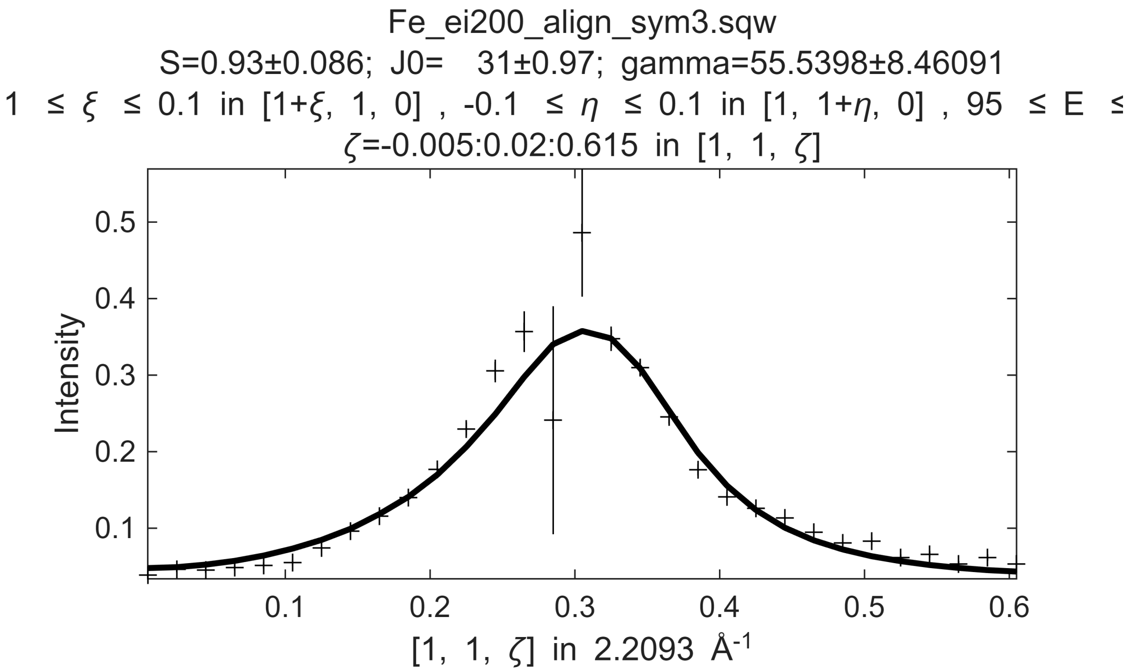
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E

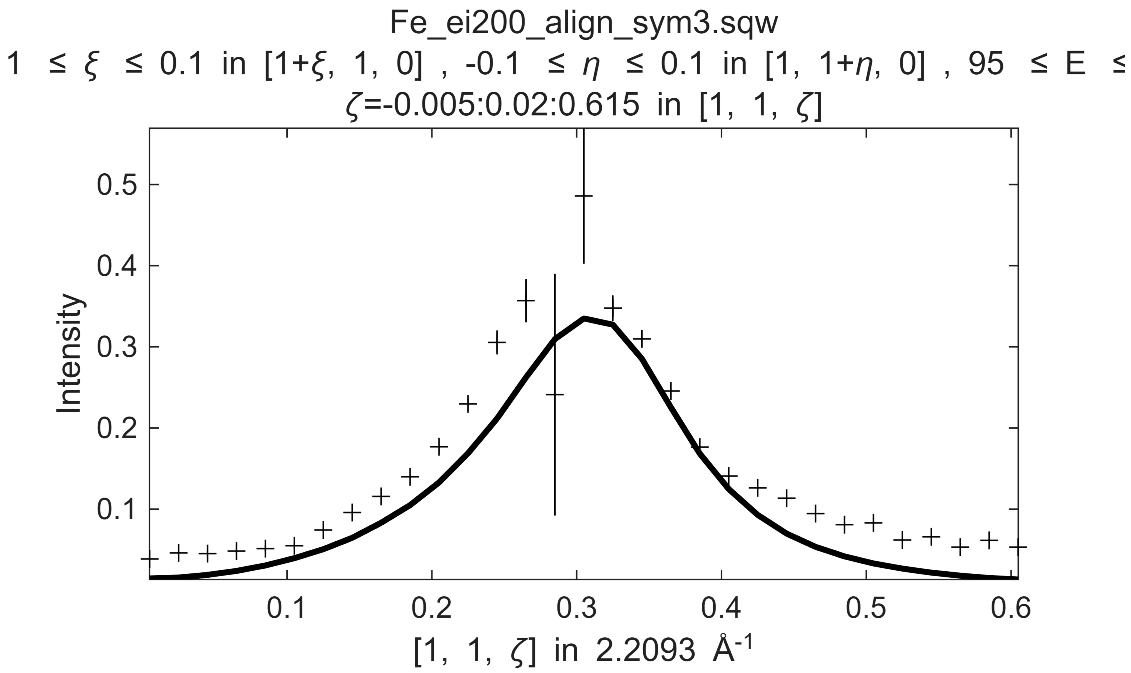
ζ=-0.005:0.02:0.615 in [1, 1, ζ]



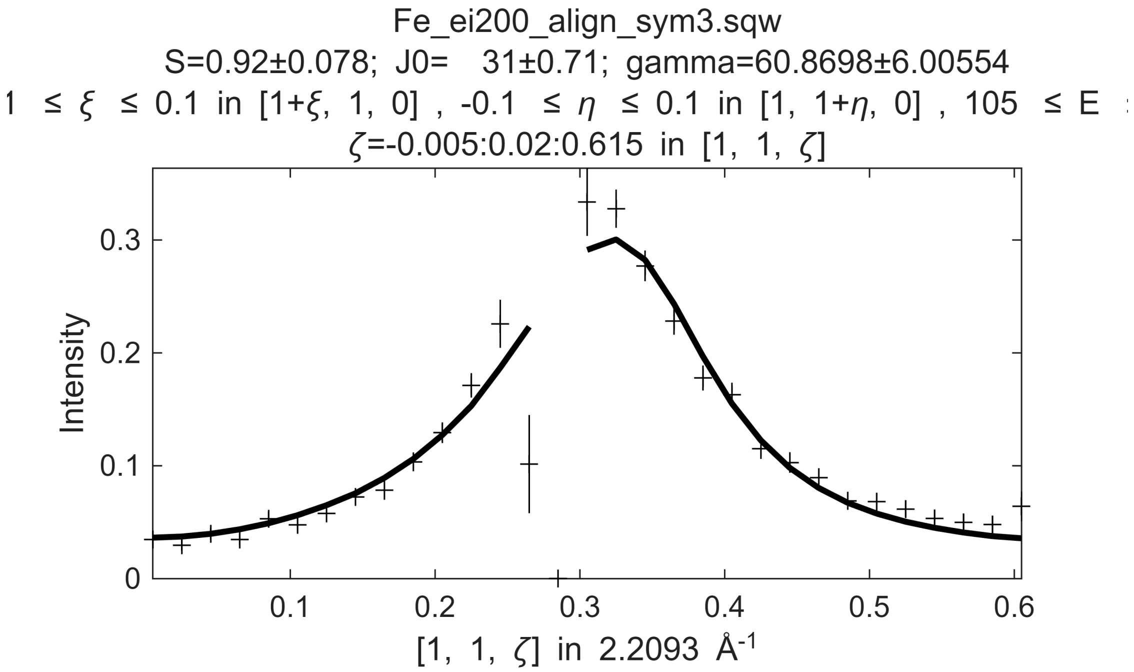

Fe_ei200_align_sym3.sqw
.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
0.5
0.4
0.3
0.2
0.1
0.1
0.2
0.3
0.4
0.5
0.6
[1, 1, ζ] in 2.2093 Å-1

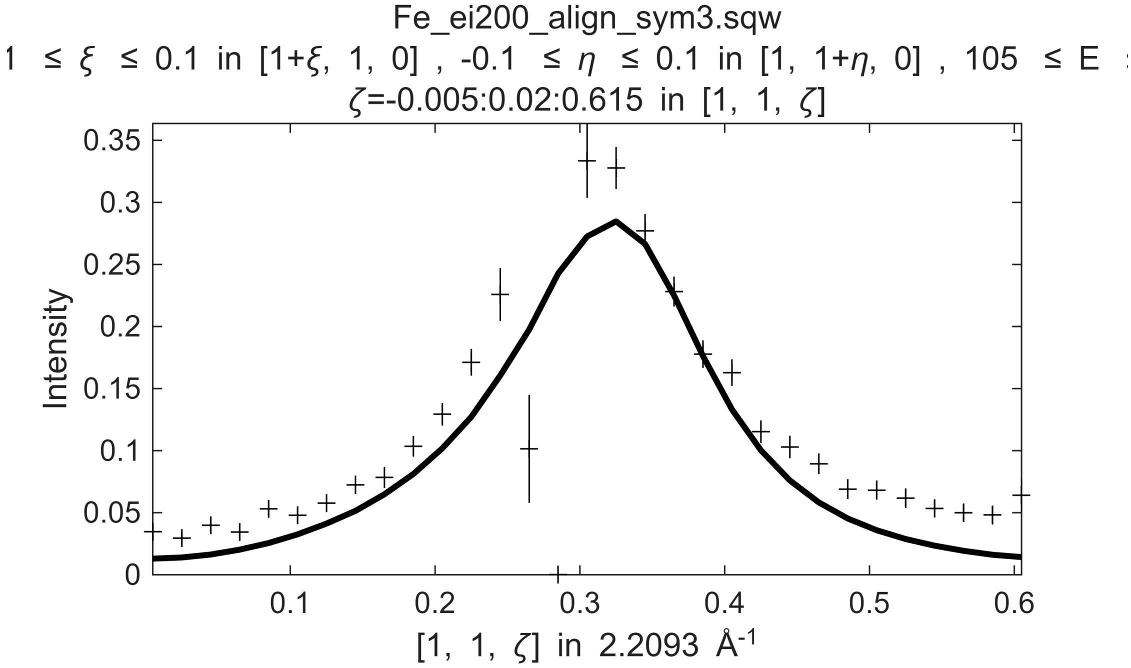
****
**** En = 100±5
****

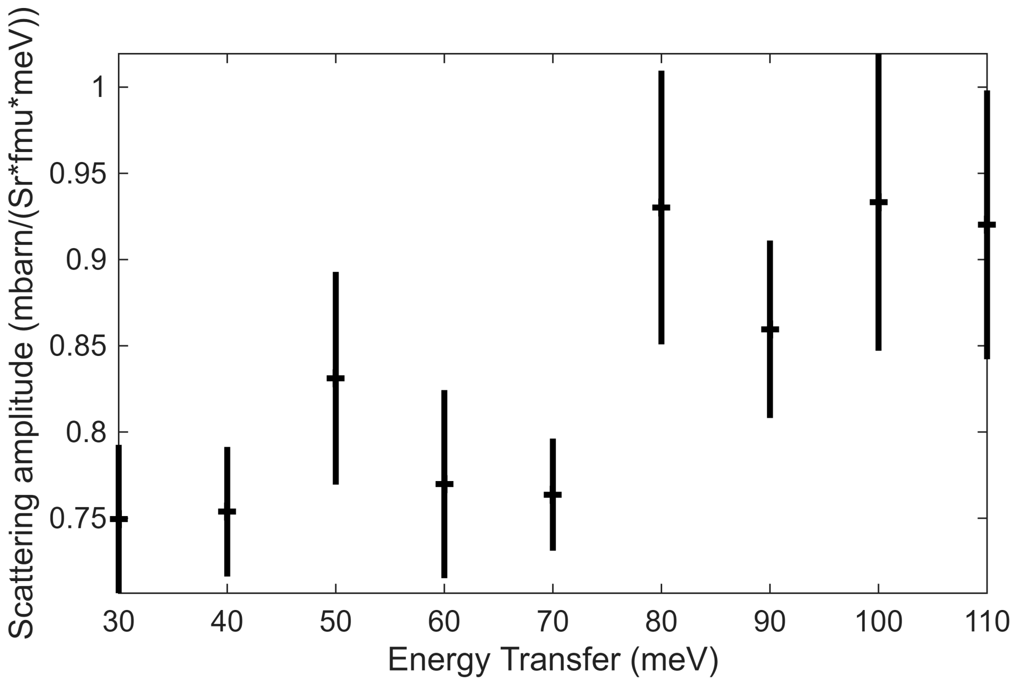

Fe_ei200_align_sym3.sqw
S=0.93±0.086; J0= 31±0.97; gamma=55.5398±8.46091
1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 95 ≤ E ≤
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
0.5
0.4
0.3
0.2
0.1
0.1
0.2
0.3
0.4
0.5
0.6
[1, 1, ζ] in 2.2093 Å-1

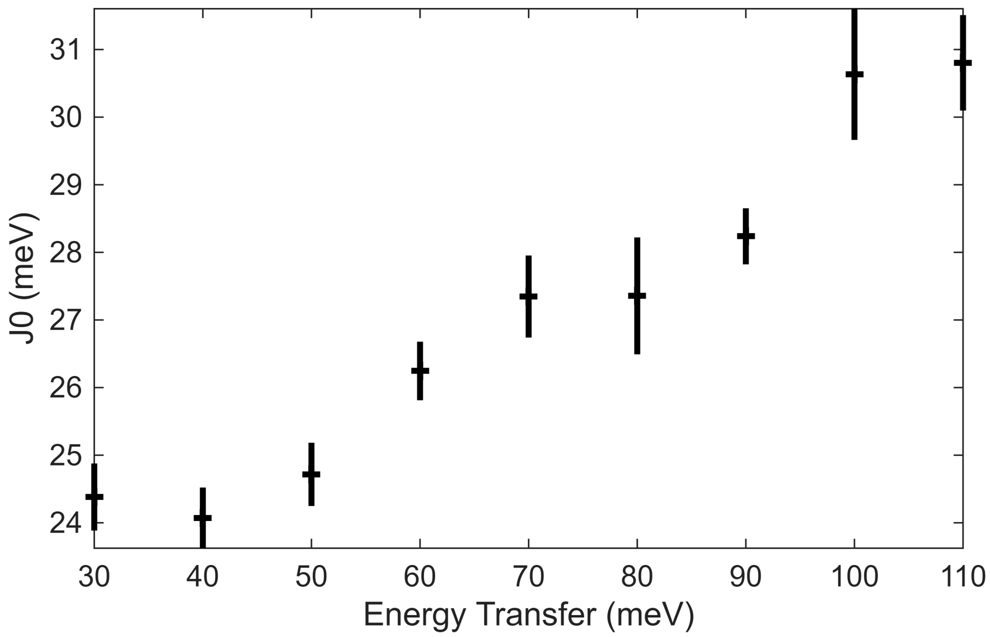



```
*******************************
**** En = 110±5
*******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 3.2258
```

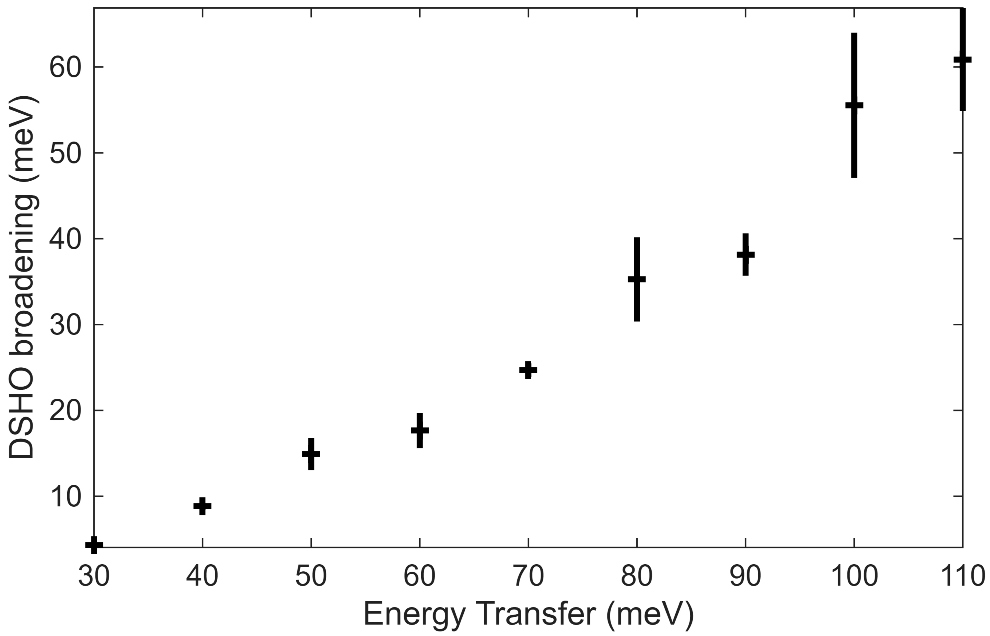



Fe_ei200_align_sym3.sqw
1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 105 ≤ E
ζ=-0.005:0.02:0.615 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0.1
0.2
0.3
0.4
0.5
0.6

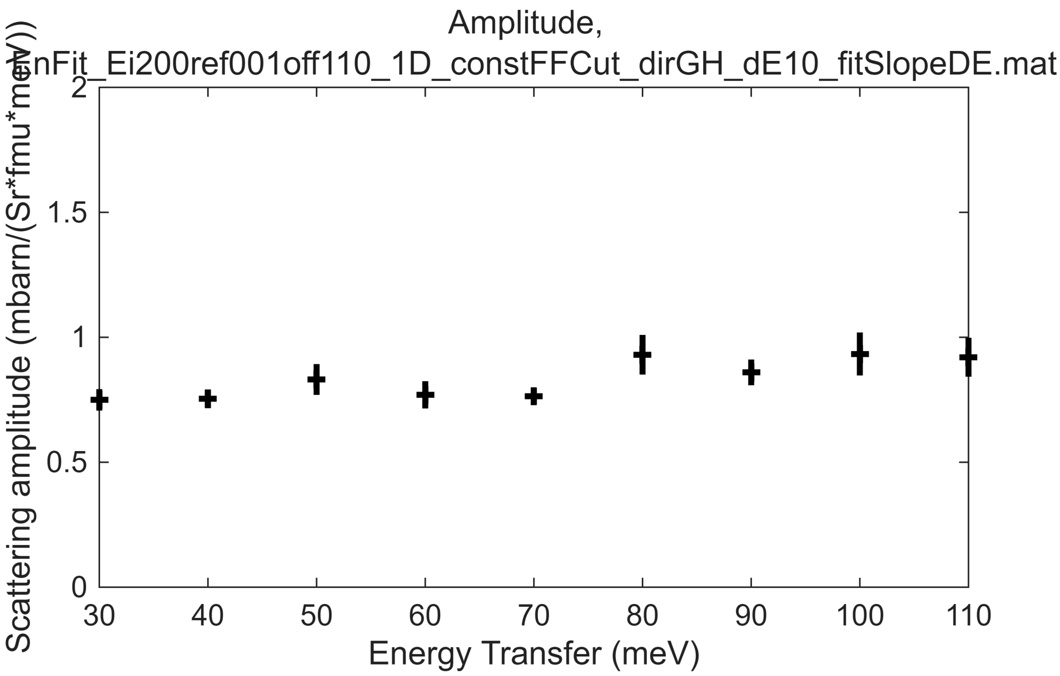

Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
1
0.95
0.9
0.85
0.8
0.75
30
40
50
60
70
80
90
100
110

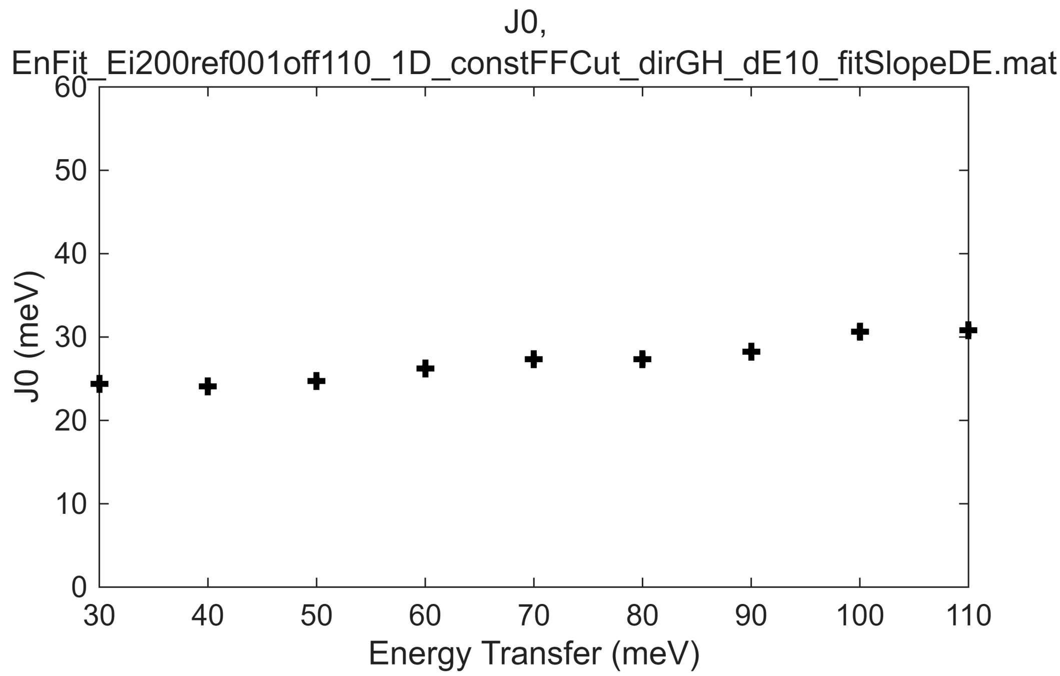


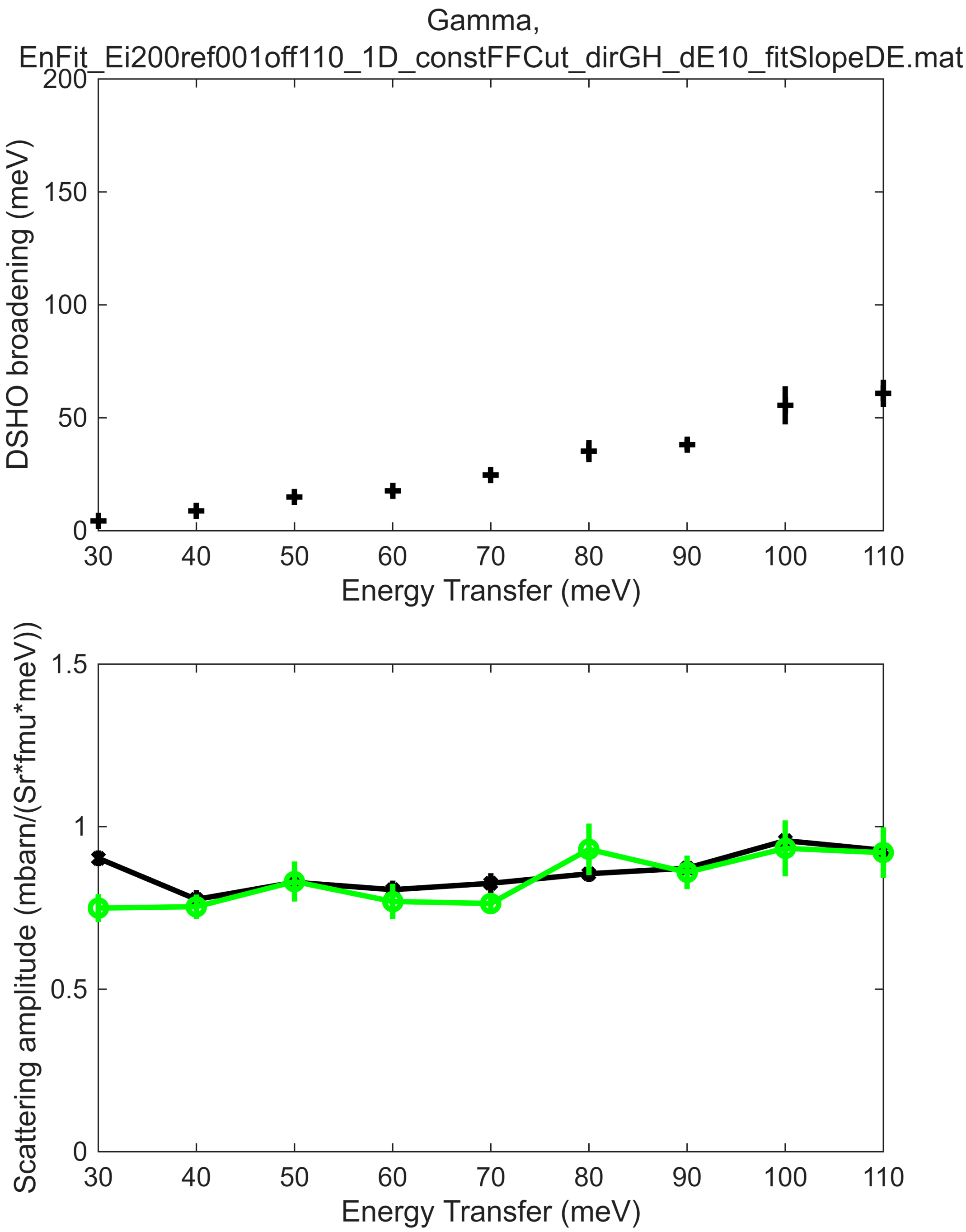
J0 (meV)
Energy Transfer (meV)

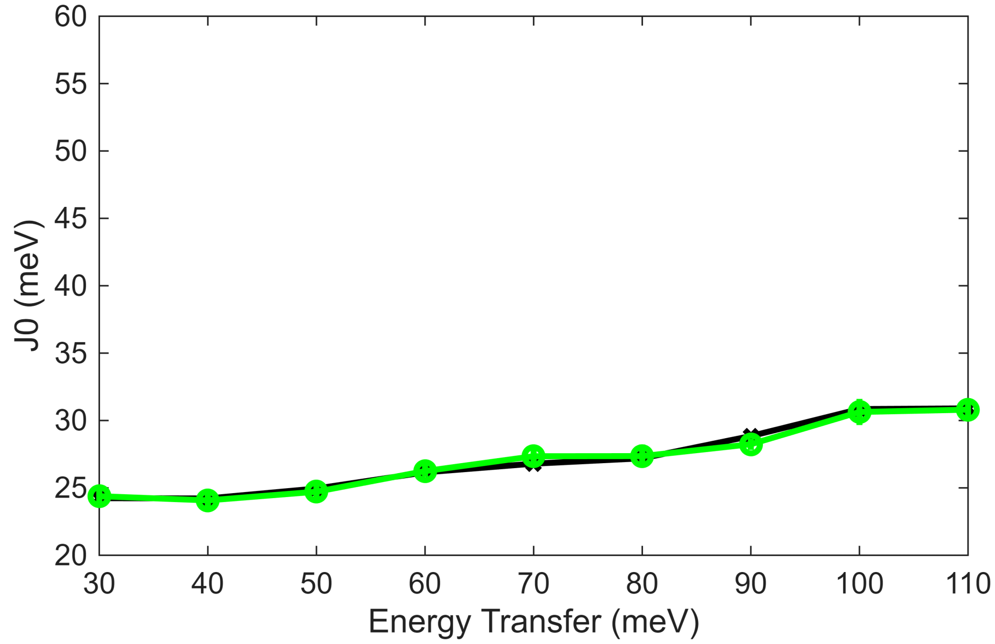


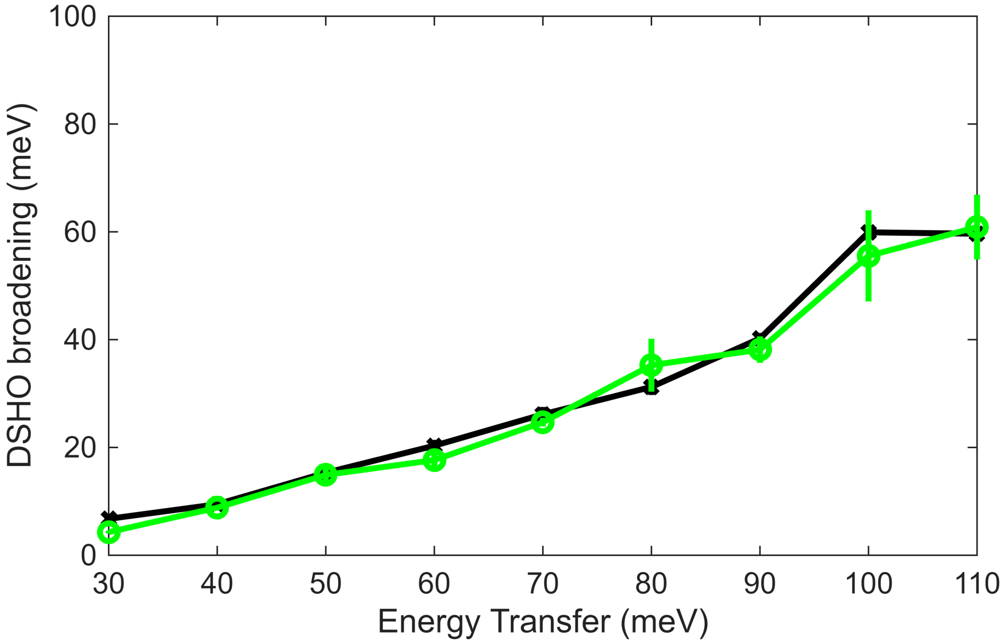
DSHO broadening (meV)
Energy Transfer (meV)


Amplitude,
EnFit_Ei200ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
0
0.5
1
1.5
2
30
40
50
60
70
80
90
100
110


J0,
EnFit_Ei200ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
J0 (meV)
Energy Transfer (meV)
0
10
20
30
40
50
60
30
40
50
60
70
80
90
100
110

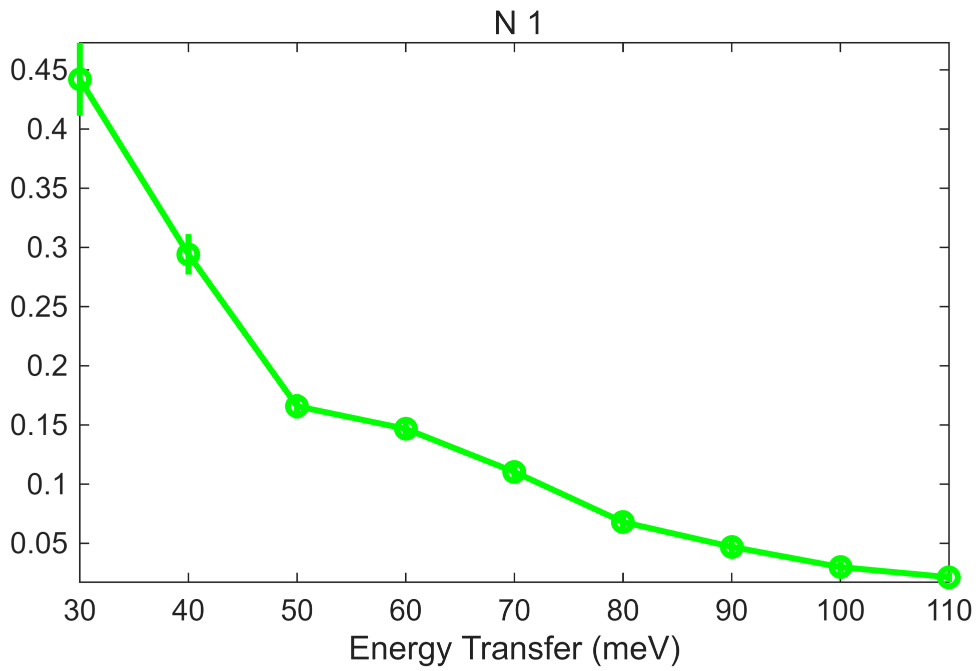

Gamma,
EnFit_Ei200ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
DSHO broadening (meV)
200
150
100
50
0
30
40
50
60
70
80
90
100
110
Energy Transfer (meV)
Scattering amplitude (mbarn/(Sr*fmu*meV))
1.5
1
0.5
0
30
40
50
60
70
80
90
100
110
Energy Transfer (meV)



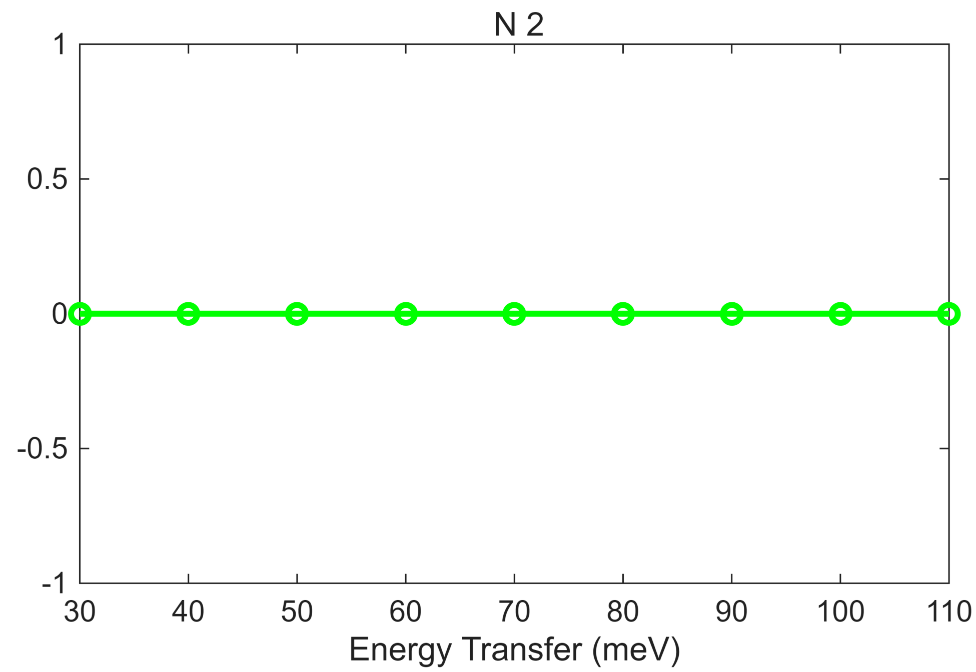



```
beep
beep
beep
```

```
bg_data = extract_bg_par(fit_res_rec.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure;
```




```
pd(bg_data(2));lx(fit_res_rec.all_fit_par(1).en,fit_res_rec.all_fit_par(end).en);
keep_figure
```




```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure
```

```
```

```
fit_resEi200 = fit_res_rec;
save(res_name,"fit_resEi200")
```